AHMAD ESSAMELDIN AHMAD

Mechanical Design Engineer — Robotics & Industrial Automation

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linkedin.com/in/ahmad-esaameldin-ahmad-b266321b3 | Cairo, Egypt | Relocating to the U.S. — Authorized to work without sponsorship

SUMMARY

Mechanical Design Engineer with 2+ years specializing in robotics and industrial automation. Experienced in full-cycle mechanical design — from concept to operational deployment — including leading the ground-up development of a new robotic platform at ARIA Technologies that drew company-wide recognition after successful operation. Proficient in SolidWorks, Fusion 360, and hands-on integration of KUKA, Yaskawa, Mitsubishi, and Nachi robots. I don't design to check a box — I design systems that are technically sound and visually compelling. Authorized to work in the U.S. without sponsorship and actively seeking relocation opportunities in robotics and automation.

EXPERIENCE

Mechanical Design Engineer | **ARIA Technologies** *Aug 2023 – Present*

Cairo, Egypt | Promoted: Intern → Junior Mechanical Design Engineer → Mechanical Design Engineer

- Identified a critical structural inefficiency in ARIA's cartesian robot line — all axes were built from heavy, expensive steel — and independently initiated an R&D program to redesign the Z-axis using custom aluminum extrusion profiles, applying topology optimization and design study tools to match original structural integrity at significantly lower weight and cost.
- Extended the aluminum redesign to the X-axis after a full year of R&D, including external consultant validation; identified and engaged Dellemar as the extrusion partner, prepared and delivered a technical presentation to their team, and secured an RFQ — resulting in successful deployment across the robot line.
- Built and led the mechanical design department from the ground up, growing the team to 4 design engineers and interns while managing workload distribution, design reviews, and delivery timelines.
- Represented ARIA Technologies at international exhibitions in Saudi Arabia, handling both on-site robot installation and live technical presentations to clients and industry visitors.
- Conducted in-person meetings with international suppliers in Malaysia, China, and Saudi Arabia, and collaborated with the pre-sales department to deliver conceptual designs and proof-of-concept solutions for client proposals.
- Transitioned component manufacturing from outsourced fabrication to in-house CNC machining, reducing lead time and procurement costs by 30%.
- Built from scratch a Notion-based pipeline tracker for the design department, improving project visibility and delivery coordination across R&D, electrical, and production teams.
- Maintained 2D and 3D engineering drawings to ISO/ASME standards, applying GD&T, DFM, and DFA principles using SolidWorks and Fusion 360 across all active projects.

TECHNICAL SKILLS

CAD & Design: SolidWorks (CSWP), Fusion 360, Autodesk Inventor, Blender
Simulation & Analysis: MATLAB, Simscape, FEA, KUKA Process Simulate, Motoman (Yaskawa) Simulation
Visualization & Rendering: KeyShot, DaVinci Resolve, Canva
Manufacturing & Standards: DFM, DFA, GD&T, ISO/ASME Drawing Standards, CNC Machining
Robot Brands: KUKA, Yaskawa, Mitsubishi, Nachi
Productivity & Planning: Notion, Microsoft Excel, Technical Presentations

SELECTED PROJECTS

ARIA Technologies — Industrial Projects

Secondary Packaging & Robotic Palletizing System *(Battery Manufacturing Facility)*

Served as Mechanical Design Lead with full project ownership. Designed and deployed a complete automated secondary packaging and robotic palletizing cell integrating the ARIA P60 articulated robot, a custom one-axis linear servo-driven robot, conveyor systems, and perimeter safety enclosures. System engineered for 24/7 continuous operation in a chemically harsh battery manufacturing environment. Responsibilities included system architecture, layout planning, component selection, shop floor assembly supervision, and on-site installation and commissioning.

End-of-Line Robotic Palletizing System *(MAFI — Egyptian Client)*
Designed a fully automated end-of-line palletizing solution. Independently sourced and engaged a Swedish EOAT manufacturer — prepared technical requirements, initiated contact, received design proposal and RFQ. Also designed a secondary EOAT integrated with the ARIA robot for automated empty pallet feeding to the line.

Academic Projects

Flight Simulator Motion Cueing Platform & Active Pilot Controls *(Graduation Project)*
Designed and built a full 6-DOF motion cueing platform with active pilot controls, implementing an LQR-optimized washout filter algorithm for real-time motion simulation. Won First Place at the 16th UGRF national research forum.

EMG-Based Prosthetic Arm with Force Feedback
Developed a myoelectric prosthetic arm controlled via EMG signals with integrated force feedback for intuitive grip modulation.

AI Real-Time Hand Gesture Prosthetic Arm
Designed a prosthetic arm driven by real-time computer vision and AI-based hand gesture recognition, enabling electrode-free natural control.

Computer Vision Lane-Keeping Mobile Robot
Built an autonomous mobile robot using computer vision for real-time lane detection and steering correction.

3D Printer — Design & Manufacturing
Designed and manufactured a fully functional 3D printer from scratch covering mechanical frame, motion system, and full assembly.

EDUCATION

B.S. in Mechatronics Engineering — MSA University, Giza, Egypt
Sep 2018 – Jul 2023 | GPA: 3.94 / 4.00 | Ranked 2nd in cohort
Graduation Project: Flight Simulator Motion Cueing Platform & Active Pilot Controls System

CERTIFICATIONS

- Certified SolidWorks Professional (CSWP) — Dassault Systèmes
- Certified SolidWorks Associate (CSWA) — Dassault Systèmes

AWARDS & HONORS

- Top Performer 2024 — ARIA Technologies
- First Place & Honor Shield — 16th Undergraduate Research Forum (UGRF), Nile University & Egyptian Knowledge Bank
- Academic Excellence — Ranked 2nd across all years of study, Mechatronics Engineering cohort, MSA University

PUBLICATIONS

- Designing an Optimal Washout Filter for a Flight Simulator Using Linear Quadratic Regulator (LQR) — Arabian Journal for Science and Engineering (Springer) — Under Review
- Flight Simulator Motion Cueing Platform: Interfacing and Pilot Controls Prototyping — MSA Engineering Journal, Feb 2023 — Published

EXHIBITIONS & INDUSTRY EVENTS

- Industrial Transformation by Hannover Messe 2025 — Riyadh, KSA (ARIA Technologies representative)
- LEAP 2025 — Riyadh, KSA (ARIA Technologies representative)
- WAM Saudi 2026 — Saudi Arabia (ARIA Technologies representative)
- 7th Cairo Innovation Exhibition 2022 — Cairo, Egypt (University representative)
- MTC 2023 — Egypt (University representative)

LANGUAGES

Arabic (Native) **English** (Proficient)